# One-Sided Flow Becomes Toxic When the Market Moves With It

Mark Aron Szulyovszky (Aperiodic) — LinkedIn post, 1 October 2026
https://www.linkedin.com/feed/update/urn:li:activity:7511449664941064192/

One of our most interesting metric is `flow toxicity`:

It measures how likely it is that the flow hitting the book knows something. The idea comes from VPIN, from Easley, López de Prado and OHara (2012). VPIN measures order imbalance in volume time and relates it to liquidity providers pulling back. Our "flow toxicity score" is a simpler measure on ordinary time bars.

It tries to identify the moments when aggressive order flow becomes difficult for the market to absorb. One-sided trading alone isn't necessarily interesting, and neither is volatility by itself, although of course correlated. Flow toxicity helps identify when both become unusual at the same time.

We define it as "past-normalized signed notional flow multiplied by past-normalized short-horizon realized volatility".

We serve it at Aperiodic, alongside the 100+ PIT market microstructure, flow & liquidity metrics.

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# One-sided flow becomes toxic when the market moves with it.

Inspired by VPIN (Easley, López de Prado and O'Hara, 2012), a simpler flow toxicity measure that works with regular time bars.

